

A PROJECT TITLE
ON
Traffic Congestion Level Prediction Using Random Forest and Gradient-Boosted Trees
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

Roll Number(s)	Name of the Student
2520030592	PUDOTA ASRITHA CHOWDARY
2520030594	PYDI DEVI SREE
2520030600	CHENNU GAYATRI SRI HARSHITA

Under the Esteemed guidance of

GADDAM
RAVINDRA SIR
CSE DEPARTMENT



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING.

Name of Title: Traffic Congestion Level Prediction Using Random Forest and Gradient-Boosted Trees

1. Abstract:

Traffic congestion is a major urban problem that significantly affects travel time, fuel consumption, air quality, and transportation efficiency. The continuous collection of traffic data generates large and complex datasets, making the accurate prediction of congestion levels challenging using traditional methods. This project proposes a Traffic Congestion Level Prediction System that uses supervised machine learning techniques to predict traffic congestion based on historical traffic and environmental data.

The system utilizes relevant traffic parameters, including vehicle count, average speed, road occupancy, traffic volume, time of day, weather conditions, and other related features obtained from traffic datasets. The collected data is pre-processed through data cleaning, missing value handling, feature selection, and feature scaling to improve model reliability and performance. Two machine learning algorithms, Random Forest and Gradient-Boosted Trees, are implemented and comparatively analysed. Random Forest combines multiple decision trees to produce reliable predictions, while Gradient-Boosted Trees sequentially builds trees to improve prediction accuracy by reducing previous errors.

The proposed system follows the machine learning lifecycle, including data preprocessing, model training, evaluation, and prediction. The performance of both algorithms is compared using appropriate evaluation metrics to determine the more effective model for traffic congestion prediction. By enabling the prediction of congestion levels, the system demonstrates the practical application of machine learning in intelligent transportation and supports data-driven traffic management and planning.

2. INTRODUCTION:

Traffic congestion occurs when the number of vehicles on roads exceeds the available road capacity, resulting in slower traffic movement, increased travel time, fuel consumption, and pollution. Traffic conditions can vary depending on vehicle volume, average speed, road occupancy, weather conditions, and time of day. The project aims to automatically predict traffic congestion levels using machine learning techniques. Random Forest and Gradient-Boosted Trees are used to develop and compare prediction models for identifying different levels of traffic congestion.

3. SOFTWARE REQUIREMENTS:

Python 3.x, Google Colab / Jupyter Notebook, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.

- Python 3.x – Used as the main programming language to develop the ML project.
- Google Colab / Jupyter Notebook – Used to write, run, test, and analyse the Python and ML code.

- Pandas – Used to load, clean, preprocess, and analyse the traffic dataset.
- NumPy – Used for numerical calculations and handling arrays and data operations.
- Scikit-learn – Used to preprocess data, train Random Forest and Gradient-Boosted Trees models, and evaluate their performance.
- Matplotlib – Used to create graphs and visualize traffic data and prediction results.
- Seaborn – Used to create statistical visualizations such as correlation heatmaps and model evaluation plots.

Signature of the Guide

Course Instructor